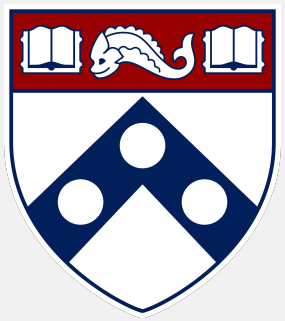
High-Arity Rademacher Complexity

joint work with Henry Towsner

Aaron Anderson

Rademacher Instructor, UPenn

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Function Classes from Formulas

We will work with a class $\mathcal{H}$ of functions $X \to [0,1]$.

Example

- Fix a structure $\mathcal{M}$ and a formula $\phi(x;y)$ (x, y can be tuples).
- Given $b \in M^y$, let $h_b : M^x \to \{0,1\}$ be the characteristic function for $\phi(M;b)$.
- Let $X = M^x$, $\mathcal{H}_\phi = \{h_b : b \in M^y\}$.

More Function Classes from Continuous Logic

In continuous logic, a formula $\phi(x)$ is interpreted over a *metric structure* $\mathcal{M}$ as a function $M^x \to [0,1]$.

Example

In a metric structure, we can let $\mathcal{H}_\phi = \{\phi^{\mathcal{M}}(x;b) : b \in M^y\}$, no extra steps needed.

Example: Randomization

- Let Ω be a probability space, and let $\mathcal{M}$ be an $\mathcal{L}$-structure.
- Let M^R be the set of random variables $\Omega \to M$.
- (Skipping measurability details.)

Definition

Make $\mathcal{M}^R$ a metric structure, the *randomization* of $\mathcal{M}$, by equipping it with relations $\mathbb{E}[\phi(\bar{x})]$ for all $\mathcal{L}$-formulas ϕ.

Theorem (Ben Yaacov, Keisler)

$\mathcal{M}^R$ has QE in this language.

Uniform Law of Large Numbers

Theorem (Vapnik-Chervonenkis)

The class $\mathcal{H}$ has finite VC-dimension if and only if

- *for every suitable measure μ on X,*
- *for all $\varepsilon > 0, \delta > 0$, there is $n \in \mathbb{N}$ such that*
- *if $\bar{x} = (x_1, \ldots, x_n)$ consists of n i.i.d. random samples from μ,*

$$\mathbb{P}_{\bar{x}}\left[\sup_{h\in\mathcal{H}}\left|\frac{1}{n}\sum_{i=1}^{n} h_b(x_i) - \mathbb{E}_\mu[h_b]\right| > \varepsilon\right] \leq \delta.$$

The original result was for sets/functions $X \to \{0,1\}$, but the real-valued version is well-known by now.

Symmetrization

Bound the average error by

- introducing another sample,
- swapping pairs x_i, y_i at random,
- decoupling with the triangle inequality.

$$\mathbb{E}_{\bar{x}}\left[\sup_{h\in\mathcal{H}}\left|\frac{1}{n}\sum_{i=1}^{n}h(x_i)-\mathbb{E}_{\mu}[h]\right|\right]$$

$$\leq \mathbb{E}_{\bar{x},\bar{y}}\left[\sup_{h\in\mathcal{H}}\left|\frac{1}{n}\sum_{i=1}^{n}(h(x_i)-h(y_i))\right|\right]$$

$$= \mathbb{E}_{\bar{x},\bar{y},\sigma}\left[\sup_{h\in\mathcal{H}}\left|\frac{1}{n}\sum_{i=1}^{n}\sigma_i(h(x_i)-h(y_i))\right|\right]$$

$$\leq 2\mathbb{E}_{\bar{x},\sigma}\left[\sup_{h\in\mathcal{H}}\left|\frac{1}{n}\sum_{i=1}^{n}\sigma_i h(x_i)\right|\right]$$

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$$\leq 2\,\mathbb{E}_{\bar{x},\sigma}\left[\sup_{h\in\mathcal{H}}\left|\frac{1}{n}\sum_{i=1}^{n}\sigma_i h(x_i)\right|\right] \leftarrow \mathcal{R}_{\mathcal{H},\mu}(n)$$

Rademacher Complexities

Given $\bar{x} \in X^n$, define the *empirical Rademacher complexity*:

$$\mathcal{R}_{\mathcal{H}}(\bar{x}) = \mathbb{E}_{\sigma}\left[\sup_{h\in\mathcal{H}}\left|\frac{1}{n}\sum_{i=1}^{n}\sigma_i h(x_i)\right|\right].$$

Taking a sup, we get the *worst-case Rademacher complexity*:

$$\mathcal{R}_{\mathcal{H}}(n) = \sup_{\bar{x}} \mathcal{R}_{\mathcal{H}}(\bar{x}),$$

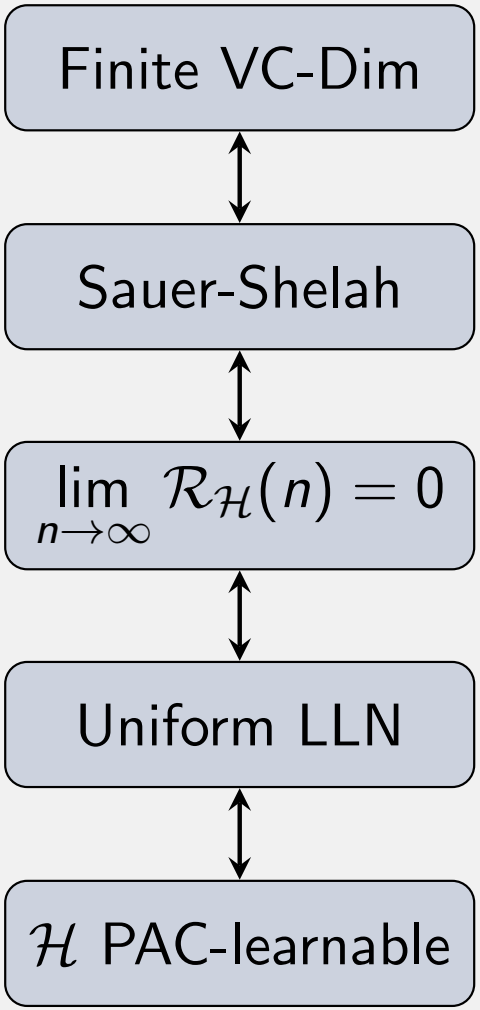
or taking an expectation, just *Rademacher complexity*:

$$\mathcal{R}_{\mathcal{H},\mu}(n) = \mathbb{E}_{\bar{x}}\left[\mathcal{R}_{\mathcal{H}}(\bar{x})\right].$$

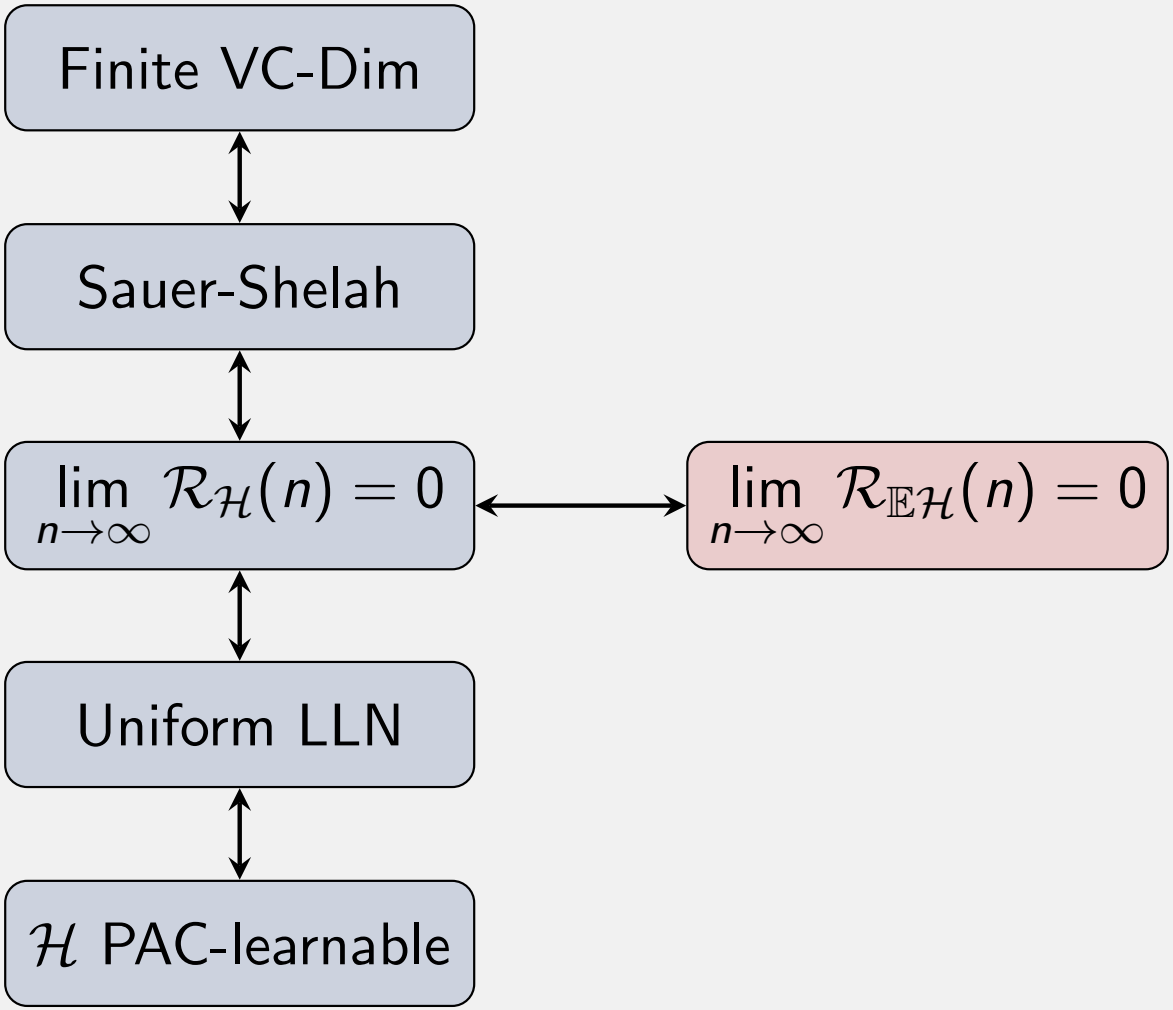
Rademacher Complexity as Discrepancy

- Let $\bar{x} \in X^n$ be a specific, worst-case, or random sample.
- Flip a coin n times to split $\bar{x}$ into two random subsamples.
- How much (in expectation) does $\frac{1}{n}\sum_{i \in E} h(x_i)$ differ between the two samples, for worst-case $h \in \mathcal{H}$?

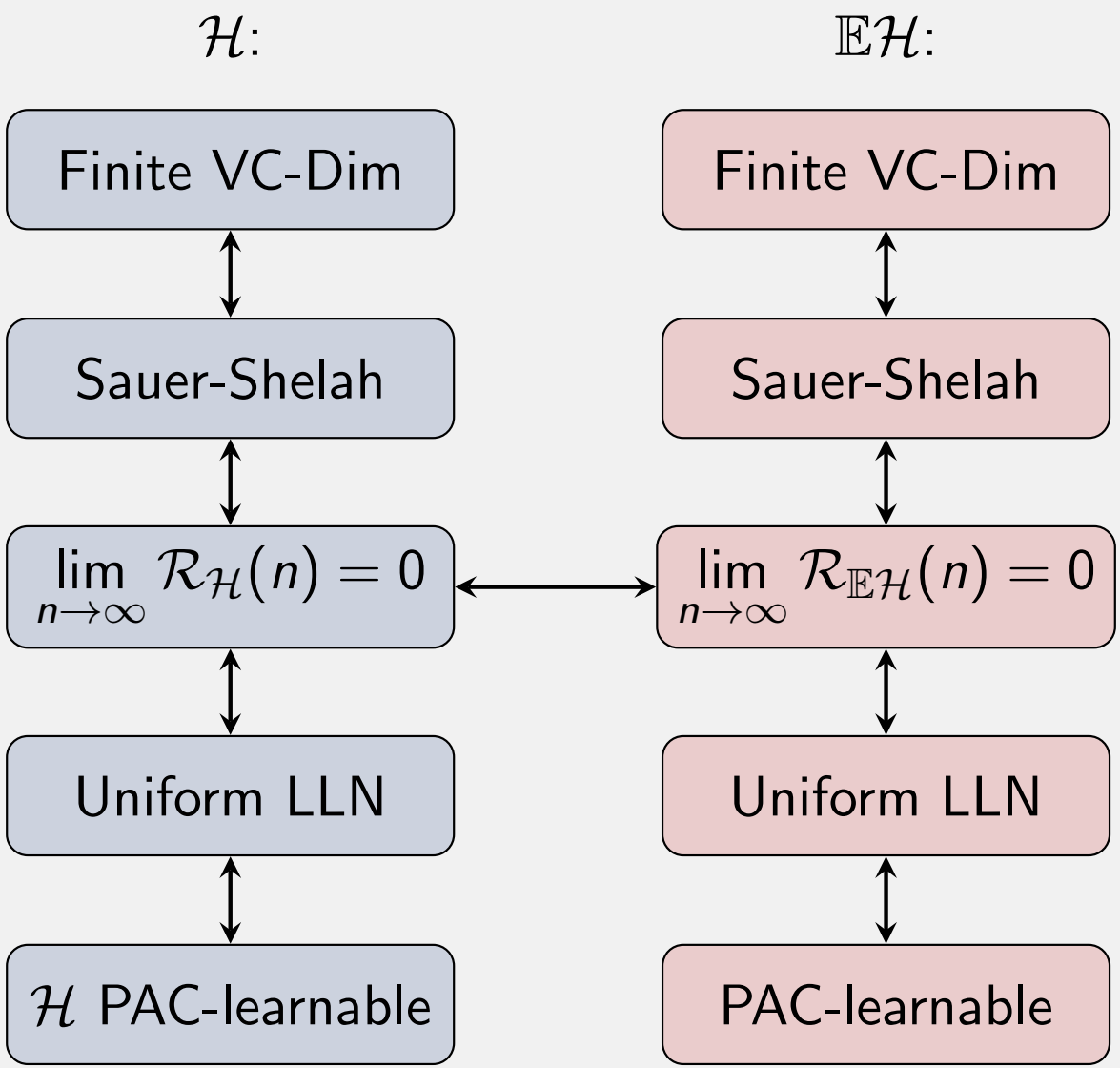
Bridge to Learnability



Bridge to Learnability and Randomization



Bridge to Learnability and Randomization



Strategy (including results and work in progress)

- Link VC-dimension (and VC_k-dimension, slicewise VC-dimension) to Rademacher complexity.
- Link stability (and slicewise stability) to *sequential* Rademacher complexity.
- Derive some statistical results.
- Show all of these are preserved under randomization.

Real-Valued NIP

Definition

For $\gamma > 0$, say $\mathcal{H}$ *γ-shatters* $\bar{x} \in X^n$ when there is a witness $v \in [0,1]^n$ such that for all $\sigma \in \{-1,1\}^n$, there is $h_\sigma \in \mathcal{H}$ with

$$\forall i \in [n], \sigma_i(h_\sigma(x_i) - v_i) \geq \frac{\gamma}{2}.$$

Definition

Say $\mathcal{H}$ has

- *γ-VC-dimension* $\geq d$ when it shatters $\bar{x} \in X^d$.
- *finite VC-dimension* when $\forall \gamma > 0$, γ-VC-dim is finite.

Definition

Say $\mathcal{M}$ is *NIP* when each $\mathcal{H}_\phi$ has finite VC-dimension.

Higher-Arity NIP

Notation

Suppose $X = \prod_{i=1}^{k} X_i$.
Given tuples $\bar{x}^i \in X_i^n$ for $1 \leq i \leq k$, define $\vec{x} = \bigotimes_{i=1}^{k} \bar{x}^i \in X^n$ by

$$\vec{x_i} = (x_{i_1}^1, \ldots, x_{i_k}^k).$$

Definition

Say $\mathcal{H}$ has

- *γ-VC_k-dimension $\geq d$* when it shatters $\bigotimes_{i=1}^{k} \bar{x}^i$ with $\bar{x}^i \in X^d$.
- *finite VC_k-dimension* when $\forall \gamma > 0$, γ-VC-dim is finite.

Definition

Say $\mathcal{M}$ is *NIP_k* when for each $\phi(x_1, \ldots, x_k, y)$, $\mathcal{H}_\phi$ has finite VC_k-dimension.

Sauer-Shelah

To bound Rademacher complexity, start with Sauer-Shelah.

Definition

Given $\bar{x} \in X^n$, let $\mathcal{H}(\bar{x}) = \{(h(x_1), \ldots, h(x_n)) : h \in \mathcal{H}\}$.

Lemma (Sauer-Shelah)

If $\mathcal{H}$ is $\{0,1\}$-valued and has VC-dim $\leq d$, $|\mathcal{H}(\bar{x})| = O(n^d)$.

Lemma (Chernikov-Palacin-Takeuchi)

If $\mathcal{H}$ is $\{0,1\}$-valued and has VC_k-dim $\leq d$, there is $\varepsilon = \varepsilon(k,d)$ with

$$\left|\mathcal{H}\left(\bigotimes_{i=1}^{k} \bar{x}^i\right)\right| = O(2^{n^{k-\varepsilon}}).$$

Sauer-Shelah as Covering Number Bound

If $\mathcal{H}$ is $[0,1]$-valued, $\mathcal{H}(\bar{x})$ is probably infinite. Bound covering numbers instead.

Definition

For $A \subseteq [0,1]^n$, let $\mathcal{N}_\gamma(A)$ be the minimum size of a γ-*cover* C:

$$\forall a \in A, \exists c \in C, |a - c|_\infty \leq \gamma.$$

Lemma (Alon, Ben-David, Cesa-Bianchi, Haussler)

If $\mathcal{H}$ has $\frac{\gamma}{4}$-VC-dimension $\leq d$,

$$\log\left(\mathcal{N}_\gamma(\mathcal{H}(\bar{x}))\right) = O\left(d(\log n)^2\right)$$

Higher Arity Covering Number Bound

Either $\mathcal{N}_\gamma(\mathcal{H}(\bar{x}))$ is small, or $\mathcal{H}$ shatters enough subtuples of $\bar{x}$ that one of them must have size d.

Theorem (Erdős)

If $I \subseteq [n]^k$ has $|I| \geq z_k(n, d+1)$, where $z_k(n, d+1) = O(n^{k-\varepsilon})$ for some $\varepsilon = \varepsilon(k, d)$, then I contains some $\prod_{i=1}^k B_i$ with $|B_i| \geq d$.

If we ask for a shattered set of size $z_k(n, d+1)$, we must shatter a *grid* of size d^k.

Lemma (A., Towsner)

If $\mathcal{H}$ has γ-VC_k-dimension $\leq d$, then for some $\varepsilon = \varepsilon(k, \gamma, d)$,

$$\log\left(\mathcal{N}_\gamma(\mathcal{H}(\otimes_{i=1}^k \bar{x}^i))\right) = O\left(n^{k-\varepsilon}\right).$$

Higher-Arity Rademacher Complexity

Definition

If $X = \prod_{i=1}^{k} X_i$, let $\mathcal{R}_{\mathcal{H}}^{k}(n) = \sup_{\bar{x}^i} \mathcal{R}_{\mathcal{H}} \left(\bigotimes_{i=1}^{k} \bar{x}^i \right)$, where $\bar{x}^i \in X_i^n$ for each $1 \leq i \leq k$.

Rademacher Upper Bounds

Theorem (variation on Massart's Lemma)

$$\mathcal{R}_{\mathcal{H}}(\bar{x}) \leq \inf_{\gamma \geq 0} \left(\gamma n + \sqrt{\frac{n}{2} \log(\mathcal{N}_{\gamma}(\mathcal{H}(\bar{x})))} \right).$$

Corollary

If $\mathcal{H}$ has VC-dim $\leq d$, then

$$\mathcal{R}_{\mathcal{H}}(n) = O\left(\gamma + n^{-1/2} \log n\right) = o(1).$$

Corollary (A., Towsner)

If $\mathcal{H}$ has VC_k-dim $\leq d$, then

$$\mathcal{R}^k_{\mathcal{H}}(n) = O\left(\gamma + n^{-\varepsilon} \log n\right) = o(1).$$

Rademacher Lower Bound

Lemma

If $\mathcal{H}$ γ-shatters $\bar{x}$, then $\mathcal{R}_{\mathcal{H}}(\bar{x}) \geq \gamma$.

Corollary (A., Towsner)

If $\mathcal{H}$ has VC_k-dim $\geq d$, then $\mathcal{R}^k_{\mathcal{H}}(d) \geq \gamma$.

Theorem (A., Towsner)

$\mathcal{H}$ has finite VC_k-dim $\iff$ $\lim_{n\to\infty} \mathcal{R}^k_{\mathcal{H}}(n) = 0$.

High-Arity Uniform Laws of Large Numbers

Using $\mathcal{R}^k_{\mathcal{H}}(n)$, we can reprove two uniform laws of large numbers, by generalizing the classic symmetrization proof.

Theorem (A., Towsner after Coregliano, Malliaris)

If $\mathcal{H}$ has bounded VC_k-dimension, for large n, samples $\bar{x}^i \in X_i^n$, and uniformly random $E \subseteq [n]^k$,

$$\mathbb{P}_E \left[\sup_{h \in \mathcal{H}} \left| \frac{1}{n^k} \sum_{\bar{i} \in [n]^k} h(\vec{x_{\bar{i}}}) - \frac{1}{|E|} \sum_{\bar{i} \in E} h(\vec{x_{\bar{i}}}) \right| > \varepsilon \right] \leq \delta.$$

In this one, the average over a subsample approximates the average over the original sample.

High-Arity Uniform Laws of Large Numbers

In this one, the average is stated in terms of conditional expectation, following the packing lemma of Chernikov, Towsner.

Theorem (A., Towsner after Chernikov, Towsner)

If $\mathcal{H}$ has bounded VC_k-dimension, for large n and any measures μ_i,

$$\mathbb{P}_{\mu_{\bar{n}^k}}\left[\sup_{h\in\mathcal{H}}\left|\frac{1}{n^k}\sum_{\bar{i}\in[n]^k} h(\vec{x}_{\bar{i}}) - \mathbb{E}[h(\vec{x}_{\bar{i}})|\mathcal{B}_{k-1}]\right| > \varepsilon\right] \leq \delta.$$

Idea: $\mathbb{E}[h(\vec{x}_{\bar{i}})|\mathcal{B}_{k-1}]$ depends only on $(k-1)$-ary information.

General Mean Widths

Definition

Given $A \subseteq [0,1]^n$, a random variable $\sigma \in \mathbb{R}^n$, let

$$w(A, \sigma) = \mathbb{E}\left[\sup_{a \in A} \sum_{i=1}^{n} \sigma_i a_i\right].$$

Can define $\mathcal{R}_{\mathcal{H}}(n), \mathcal{R}^k_{\mathcal{H}}(n), \mathcal{R}^{\text{seq}}_{\mathcal{H}}(n)$ with various A and σ.

Theorem (Rakhlin, Sridharan, Tewari)

$\mathcal{R}^{\text{seq}}_{\mathcal{H}}(n) \to 0$ if and only if $\mathcal{H}$ has finite Littlestone dimension ($\mathcal{H}$ is stable).

Slicewise NIP/Stability

Definition

If $X = \prod_{i=1}^{k} X_i$, say $\mathcal{H}$ has *slicewise VC-dimension* $\leq d$ when for all $1 \leq i \leq k$, $\bar{y} \in \prod_{j \neq i} X_i$, $\mathcal{H}(x_i, \bar{y})$ has VC-dimension $\leq d$. Similar for Littlestone dimension and real-valued versions.

Lemma (A., Towsner)

There are sequences of random variables σ defining $\mathcal{R}^{\mathrm{sl}}_{\mathcal{H}}(n), \mathcal{R}^{\mathrm{seq,sl}}_{\mathcal{H}}(n)$ which characterize finite slicewise VC-dimension, slicewise Littlestone dimension respectively.

Slicewise NIP/Stability

Corollary (Originally Coregliano, Malliaris)

$\mathcal{H}$ having slicewise VC-dimension $\leq d$ implies a uniform law of large numbers over product measures.

Theorem (A., Towsner)

$\mathcal{H}$ having slicewise Littlestone dimension $\leq d$ implies an adversarial uniform law of large numbers.

Preservation Under Randomization

Theorem (Ben Yaacov, Keisler)

If $\mathcal{M}$ is stable, so is $\mathcal{M}^R$.

Theorem (Ben Yaacov)

If $\mathcal{M}$ is NIP, so is $\mathcal{M}^R$.

Theorem (Chernikov, Towsner)

If $\mathcal{M}$ is NIP_k, so is $\mathcal{M}^R$.

Three totally different proofs!

Preservation Under Randomization

Theorem (Ben Yaacov, Keisler)

If $\mathcal{M}$ is stable, so is $\mathcal{M}^R$.

Theorem (Ben Yaacov)

If $\mathcal{M}$ is NIP, so is $\mathcal{M}^R$.
Uses Gaussian complexity, similar to Rademacher complexity.

Theorem (Chernikov, Towsner)

If $\mathcal{M}$ is NIP_k, so is $\mathcal{M}^R$.

Three totally different proofs!

Preservation Under Randomization

Lemma (A., Benedikt)

Let $\bar{\mathbf{x}}$ be a tuple of random variables in X, and σ random in $[-1,1]^n$.

$$w(\mathcal{H}_{\mathbb{E}[\phi]}(\bar{\mathbf{x}}), \sigma) \leq \sup_{\omega \in \Omega} w(\mathcal{H}(\bar{x}_\omega), \sigma).$$

Corollary

- $\mathcal{R}_{\mathcal{H}_{\mathbb{E}[\phi]}}(n) \leq \mathcal{R}_{\mathcal{H}_\phi}(n)$
- $\mathcal{R}^k_{\mathcal{H}_{\mathbb{E}[\phi]}}(n) \leq \mathcal{R}^k_{\mathcal{H}_\phi}(n)$
- $\mathcal{R}^{\text{seq}}_{\mathcal{H}_{\mathbb{E}[\phi]}}(n) \leq \mathcal{R}^{\text{seq}}_{\mathcal{H}_\phi}(n)$

Corollary

- *NIP*
- *NIP$_k$*
- *Stability*

preserved under randomization.

Future Questions

When can trace-definability dividing lines be characterized in terms of Rademacher complexity variants?
(Main missing step seems to be Sauer-Shelah.)

Connect VC_k-dimension to something closer to standard PAC learning.

These complexities are all about the *expected discrepancy* between two random subsamples. Distality/strong Erdös-Hajnal seems to be connected to other measures of discrepancy.

Thank you!